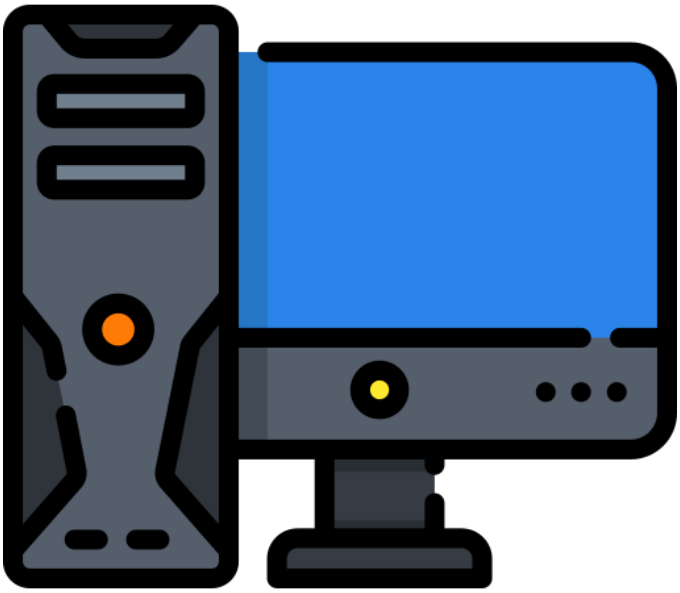
UNDER THE HOOD



OF A MODERN COMPUTER

Hardware, Software and Microbits

Name:

Class:

Mind Map Me

Emails (Microsoft Outlook)

Google Classroom

School Computer

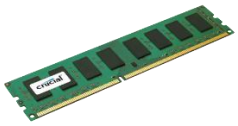


In the space above, create a mind map of all the computing devices you use at home and at school. Your mind map should include a description of how your device is used or what websites/apps you use on them.

You may find links between your devices, and you should connect these relationships in with a line between them!

Computer Breakdown Service

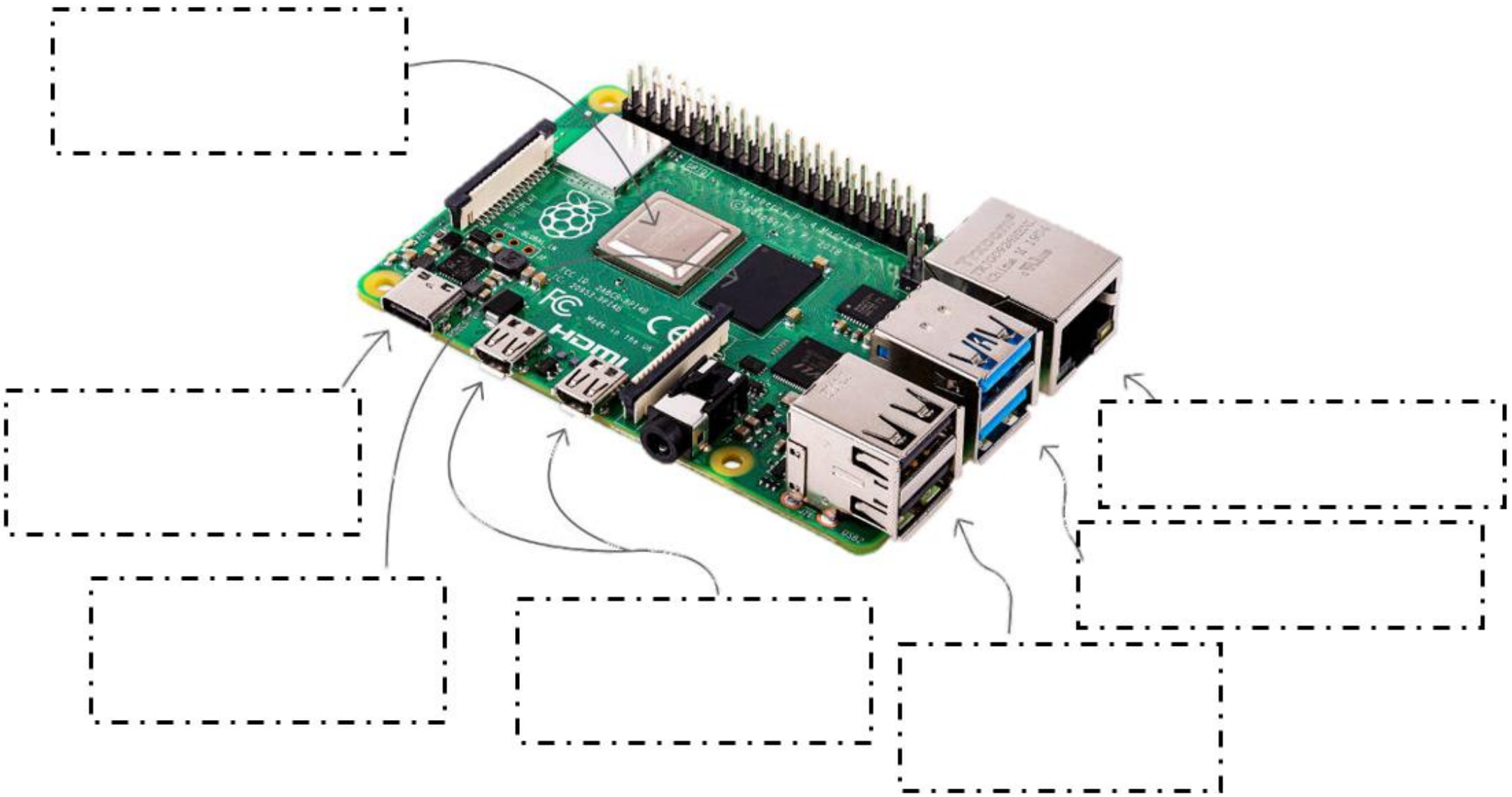
As the teacher is breaking down a computer, take note of how each of the pieces, or components, of the computer work together and its function – what it is used for and why it is needed. Some components aren't physically inside the computer, so are **external**.

Component	Function
Keyboard 	
Mouse 	
Hard Disk Drive 	
CPU 	
RAM 	
Power Supply 	

Graphics Card 	
Web Cam 	
Microphone 	
Monitor 	
Motherboard 	
Fan Unit/Heat sink 	
Chassis 	

In the space below, write something that you learnt today:

Raspberry Pi - The Pocket Sized Computer



Use The Words below to label the Raspberry Pi!

PROCESSOR | USB-C POWER SUPPLY | USB 2 | USB 3 | MICRO HDMI PORTS | ETHERNET PORT | RAM

Key Terms



Input Devices: An input device enables the user to 'input' data into a computer.

Memory: This is where a computer keeps the data that has been input, as well as software applications and the results of any processing it has carried out, for the short term. This memory is lost when the computer is turned off.

Storage Device: This is where a computer stores files that have been created, as well as software that has been installed, for the longer term.

Processor: The part of a computer that processed data according to the instructions it has been given. It provides the user with information.

Output Device: An output device enables the user to receive information from a computer.

Complete the Table below by researching as many examples of each of these devices as you can!
Note: Some are easier than others!

	Examples
Input Devices	Keyboard,
Memory	
Storage	
Processor	
Output Devices	

Spelling Test – Words to Learn.

It is important that we do not just rely on a spellchecker in order to spell key words, especially in ICT where some key concepts use an Americanised spelling or are Trademarks.

You can use multiple strategies to learn the following twenty words for a spelling test next week.

Folder

Password

Safety

Expectations

Backup

Blind Carbon Copy

Memory

Hard Disk Drive

Graphics Card

Input

Output

Processor

Hardware

Software

Motherboard

Attachment

Employee

Raspberry Pi

Eye Strain

Repetitive Strain Injury

Strategy 1

Use mnemonics - silly sentences where the first letter of each word makes up the word to be spelled. For example, "Oscar unleashed top performer under tournament".

Strategy 2

You can use look – cover – write – check where you look at the word, cover it over, write the word out and then check it in order to learn the spellings of some of the words.

Fancy WORDS

- Use the internet to research what these acronyms actually mean. After all, you use a lot of them in everyday life.
- You may also be asked in a class test what these mean, so make sure you learn them!

USB

RAM

ROM

HDD

SSD

WWW

CPU

GPU

WiFi

CD

DVD

HDMI

LAN

I/O

VGA

Games Consoles

Games consoles are very fun (and interesting in themselves!). But 'under the hood' of a games console there is very similar hardware. The most famous consoles often are very comparable, for example Sony's Playstation series and Microsoft's Xbox often have very similar hardware:



- https://www.bbc.co.uk/programmes/p07t308p
- Consoles, just like computers, understand a very simple language – just 0 and 1s.
    - That's a lot less than the letters we use in our alphabet!
- This language is known as **binary**.
- Computers read millions of these 1s and 0s every second!
- Humans program these computers, often using **variables**.

Mind Map My Console



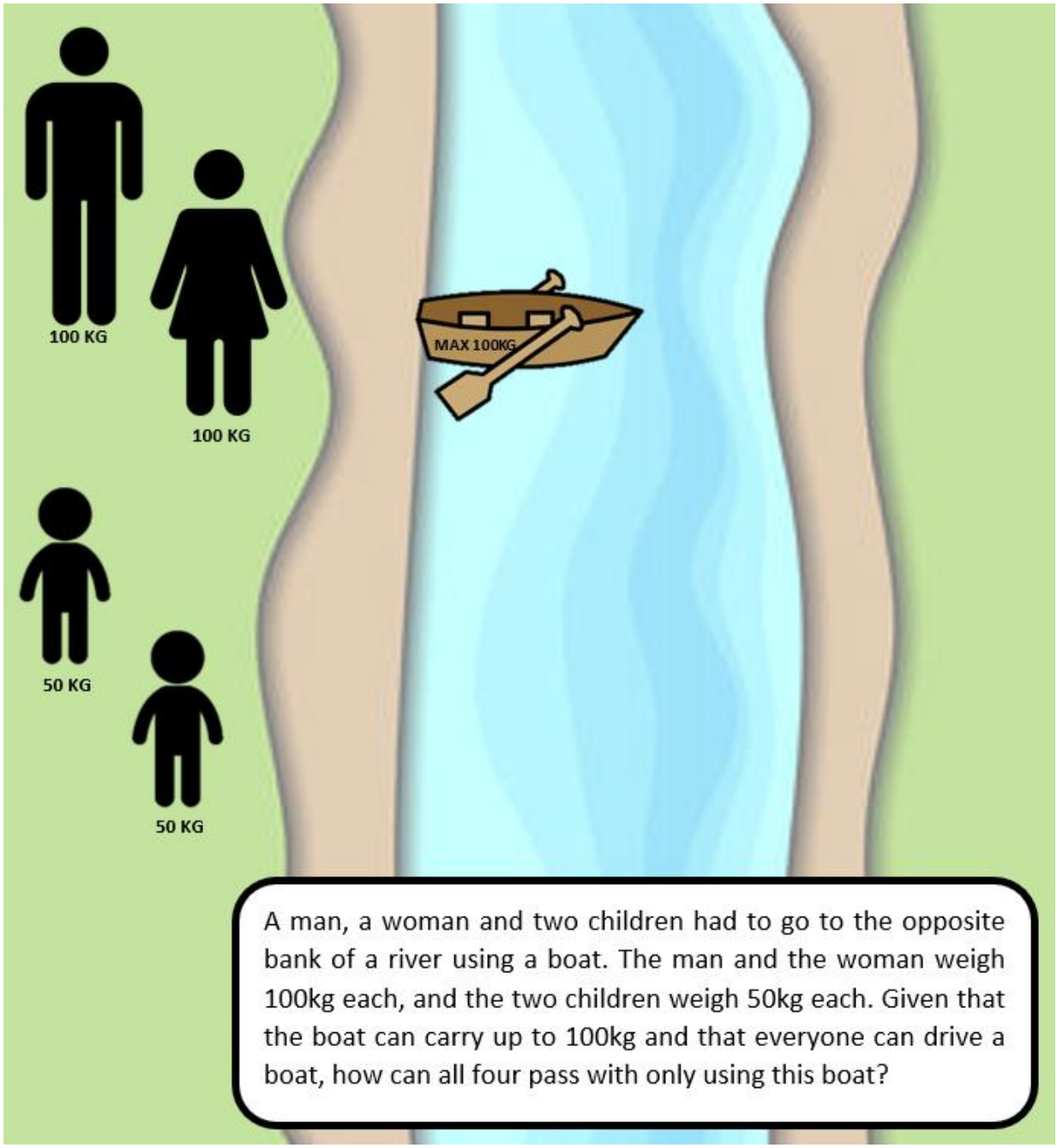
MY CONSOLE IS

_____

Create a mind map with a computer or phone or laptop or tablet or games console in the middle.

Your job is to make 'strands' to display: What each component is called, What the component is used for, A picture of the component, Are there different types/brands of this component? Is it internal or external to the computer?

The SECOND River Crossing



PAIRED TASK - Zebra Crossing Activity



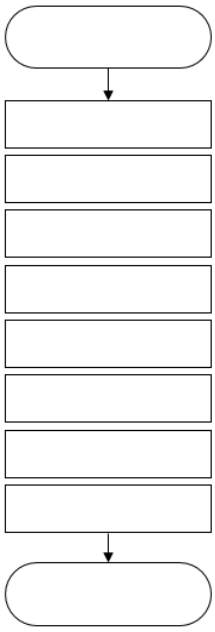
Instruction	Order
Walk to the island looking both ways and listening	
Step onto the second half of the zebra crossing	
Stop at the island	
Step onto the first half of the zebra crossing	
Stop at the side of the road looking to the other side of the Zebra crossing	1
Look right	
Look left	
Wait until traffic from the right has stopped or the road is clear	
Walk to the other side of the road looking both ways and listening	
Stay on the island until traffic from the left has stopped or the road is clear	

This is a **Zebra Crossing**. This crossing is on a wide road and has an **island** in the **middle**.

Correctly order the **instructions** to help pedestrians to use this crossing safely (the first one is done for you).

When you correctly order the **instructions**, this will create a **sequence**.

PAIRED TASK - Making a Cuppa Creating a sequence



- Stop
- Pour milk into cup
- Drink tea
- Boil water
- Add sugar
- Remove teabag from teapot
- Add water
- Pour tea into cup
- Put teabag in teapot
- Start

Look at the flowchart **instructions** and place them in order of how you would make a cup of tea.

EXTENSION:

Is there only one correct **sequence**?

What **Decisions** could you add to this flowchart?

Variable Research

Use the internet to research what a computer variable is and how it works.

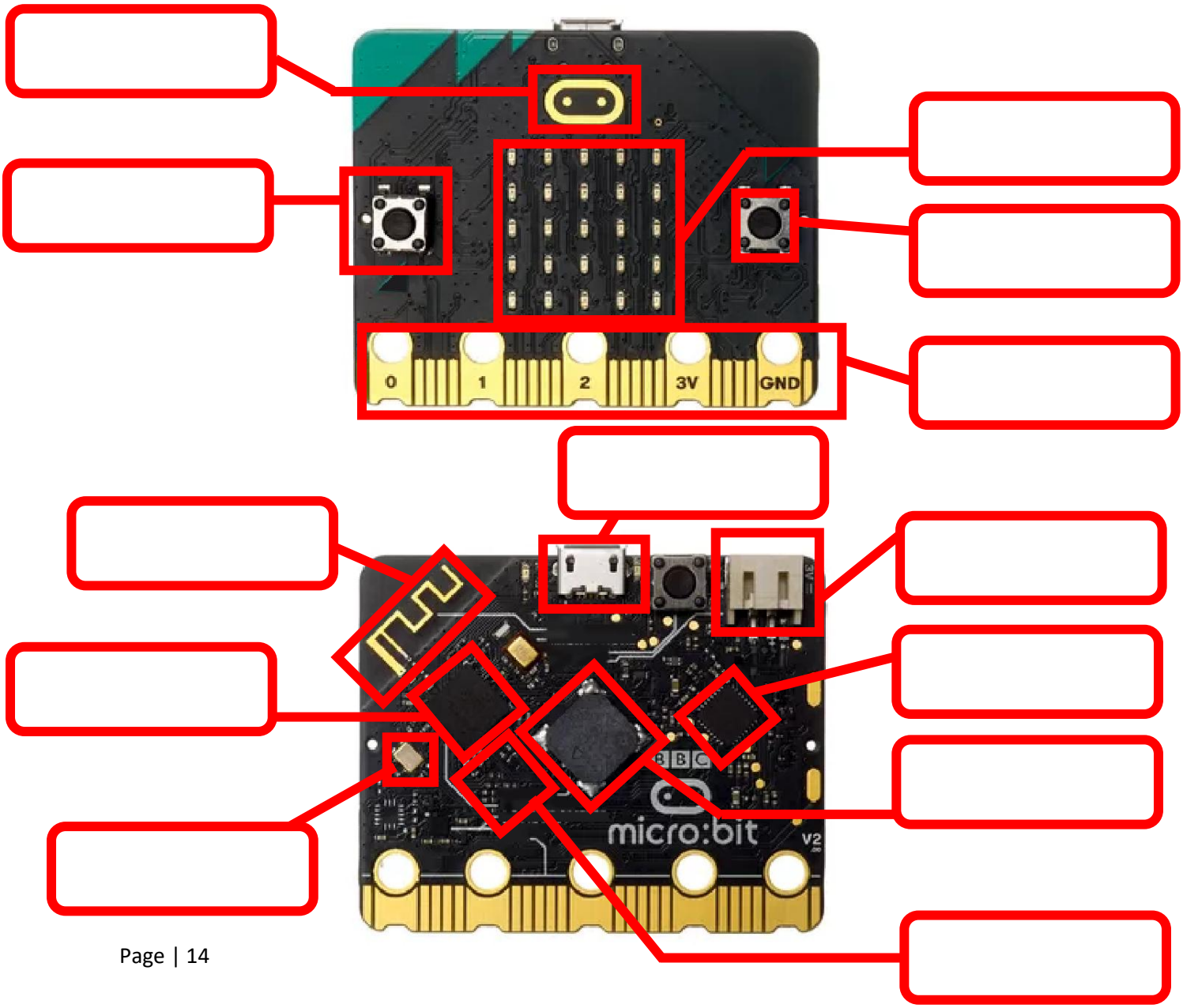
In the space below, note down your findings.

The BBC Microbit has several parts, including:

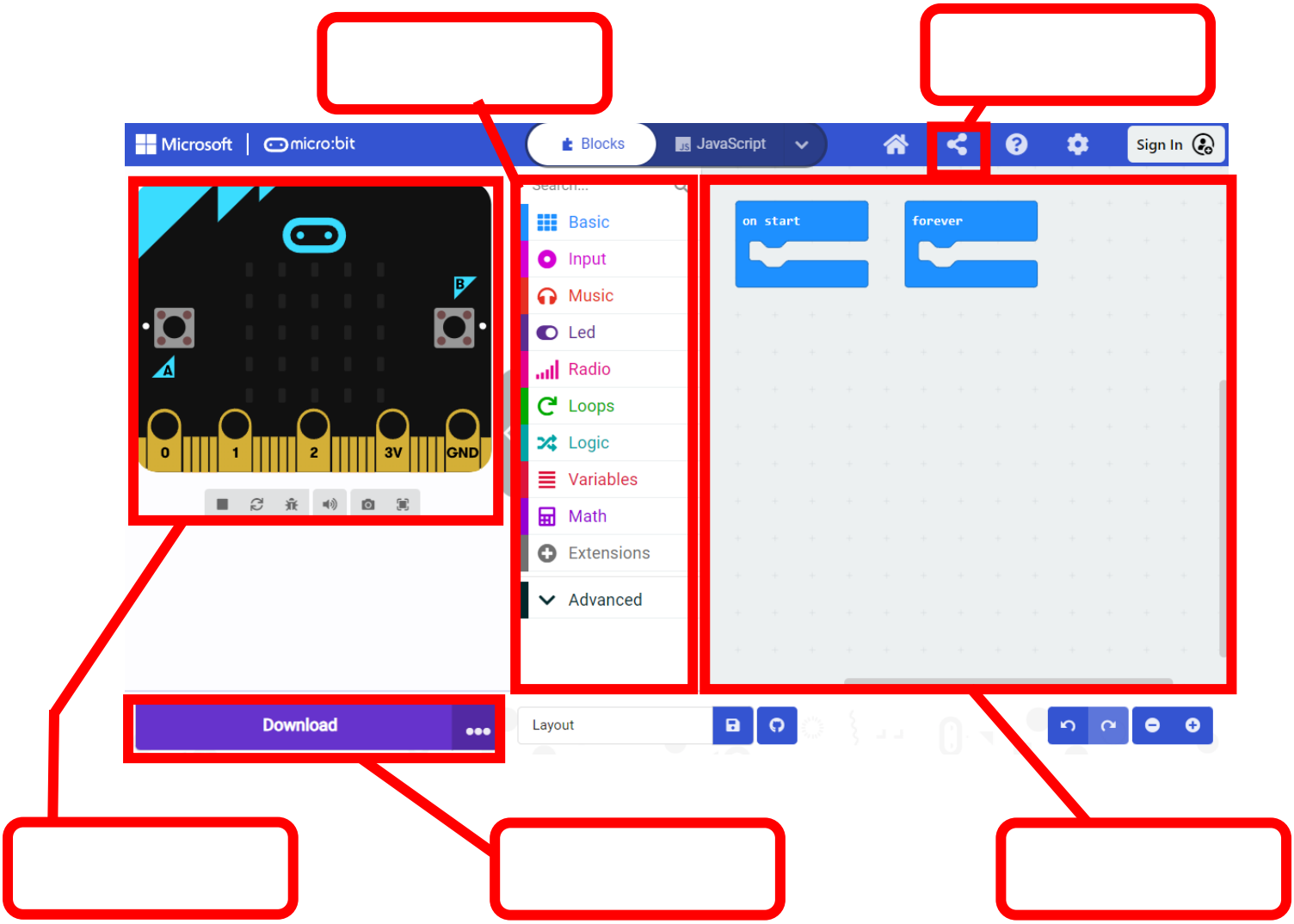
1. 25 programmable LEDs on the front of the board
2. Two programmable buttons (A and B)
3. USB connector
4. Radio & Bluetooth antenna
5. Compass
6. Battery socket
7. Accelerometer
8. Touch sensor
9. 20 pin edge connector to add further devices and sensers, and send commands to other devices

The newer models even have an on-board speaker and microphone!

25 LEDs Matrix	A Button	B Button	Microphone
20 pin edge connector	Speaker	USB connector	Battery Socket
Compass	Processor	Accelerometer	Touch-sensitive logo
Bluetooth/Radio Antenna			



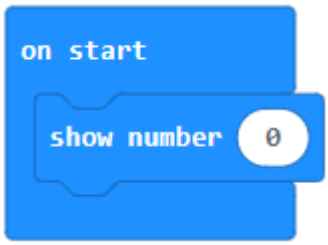
Take the tour on the Microbit website. You can click on the question mark icon at the top right and select tour if you skipped this earlier. Label the workspace below.



Basic Controls



A string is a sequence of characters, and is used to represent text. So, this 'show string' block is used to tell the Microbit to show a piece of text onto the LEDs of the Microbit.



Another really common data type is numeric data types. These usually fall into two categories:

- Integer
- Double/decimal

An integer is a whole number, and can be positive or negative. A double or decimal number had a decimal point, and can also be positive or negative.

In programming, there are other data types. For example, we use a **char** when representing a single character (letter) or symbol. A **bool** (Boolean) can be used for a true/false or on/off value and is often used to hold a 1 or a 0 value to represent one of these two options.

You will come across other data types across your programming career, but lets stick to the following:

- Integer (int)
- String
- Double
- Character (char)
- Boolean (bool)

Selection

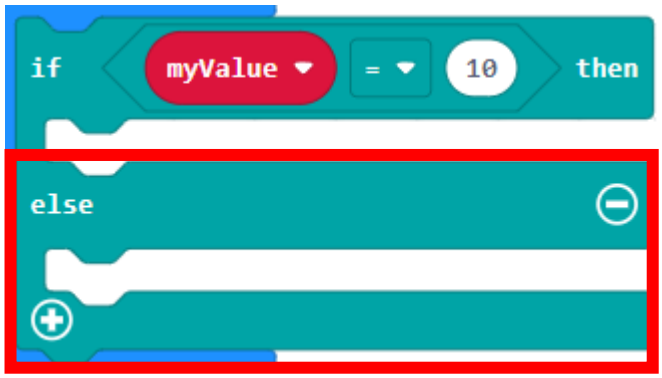
When programming, we use logic gates to decide if a piece of code should run or not. This is a programming concept of **selection**, which uses an **if** statement.



A piece of code will only run **if** a condition is true.

IF statements are usually paired with **else** statements.

This means that if a condition **is not true**, then you can make something else happen.



Loops

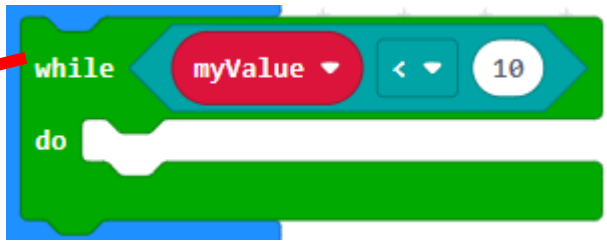
One of the most common structures we use in programming is loops, or more commonly known as **iteration**s. This allows certain code to be repeated without having to retype it over and over again. There are two types of loops;

- Condition controlled loops
- Count controlled loops

Condition Controlled Loops

Condition-controlled loops, also known as "while loops" or "do-while loops," are structures in programming that repeatedly execute a block of code as long as a specified condition is true.

The condition is evaluated before each iteration. If the condition is true, the loop body is executed. After the execution of the loop body, the condition is checked again. If the condition is false, the loop terminates.

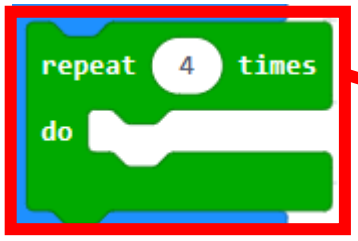


We don't know how many times this is going to loop, as we have to keep repeating until the value of myValue changes to be above 10

Count Controlled Loops

Count-controlled loops, also known as "for loops" or "foreach loops," are structures in programming that iterate a specific number of times.

Count-controlled loops are used when a specific number of iterations is required, or when iterating over elements of a collection or sequence.



We know immediately that this code will repeat 4 times.